

Chinmay Kapoor

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Applied Data Scientist | ML & LLM Systems Engineer | Enterprise Analytics & Decision Intelligence

SUMMARY

Applied Data Scientist & AI/ML Systems Engineer with experience building end-to-end, production-ready systems for analytics, forecasting, & agentic AI workflows. Strong background in Python, JavaScript, SQL, FastAPI, cloud/local deployment, Git/GitHub CI/CD, model evaluation, feature engineering, observability, & modular data/ML pipelines. Published research across spatio-temporal modeling, time-series forecasting, & applied machine learning, with hands-on systems experience in financial profiling, fraud detection, stock forecasting, RAG & hybrid deterministic + LLM workflows.

EXPERIENCE

Field Application Engineer - AI Specialist, MIMIK (Vancouver, BC) Oct 2025 - Present

- **HybridTrax + Cognifix:** Worked on real-time incident detection and agentic AI workflows where network, device, and location signals had to be converted into reliable triggers for local AI agents and response systems.
- Built ML-backed Python watcher services and signal-correlation pipelines for BLE beacon/location data, improving how incoming events were captured, processed, and passed into downstream model-triggered workflows.
- Built rapid POCs around emerging AI and Edge AI use cases, combining TTS, audio recognition, computer vision models, REST scripts, and lightweight prototype interfaces to show how new capabilities could fit into MIMIK's architecture.
- Tested trigger-response behavior using A/B testing, fault injection, load testing, and edge-case scenarios, helping identify cascading failures and improving incident-handling accuracy by 30%.
- Supported agentic AI workloads by connecting deterministic service logic with model-triggered actions, including computer vision and audio recognition components deployed in local AI environments.
- **Blackbox View:** Built tracing workflows to follow API calls, service behavior, execution paths, and runtime bottlenecks across distributed AI applications, helping the team debug failures and explain system behavior during deployment.
- Created d3.js-based codegraph and system-flow views to visualize service dependencies, data movement, and pipeline structure during architecture and debugging discussions.
- Analyzed load patterns, delayed responses, and trigger failures in high-access environments, using runtime traces and data-flow checks to isolate bottlenecks and improve production reliability.

Data Scientist - Lead, Swissvault Global (Vancouver, BC) Jan 2024 - Oct 2025

- **COMPIO:** Built a business intelligence and financial profiling platform that collected company, market, news, and financial data from APIs and web sources, then generated structured reports using Python, FastAPI, RAG, vector search, S3 storage, and document-generation workflows.
- **SVSI:** Developed a secure system-support RAG powered interface for querying private enterprise data stored in VaultFS, with focus on document ingestion, embeddings, vector search, model response quality, and privacy-aware local/cloud OpenAI compatible LLM endpoint API.
- Supported model-choice decisions by analyzing the actual data pattern first, including telemetry trends for forecasting, failure signals for classification, retrieval quality for RAG workflows, and privacy constraints for local LLM deployment.
- Designed forecasting and predictive-maintenance workflows for storage and system-performance use cases, including telemetry analysis, hardware health indicators, trend detection, and early signals of storage degradation or workload instability.
- Applied time-series modeling concepts to long-running system and telemetry data, using temporal features, lag behavior, rolling-window patterns, and long-term dependency analysis to support operational monitoring and capacity-planning decisions.
- Built batch and streaming ETL pipelines for semi-structured telemetry and system performance data, improving transformation reliability and reducing load/processing time by over 30%.
- Worked on data access strategies for large SQL-backed and file-backed datasets, focusing on efficient retrieval paths, caching, reduced repeated reads, and reliable query behavior for analytics and AI-assisted workflows.
- Worked directly with clients and internal software, sales, hardware, and data engineering teams to explain data pipeline architecture, resolve deployment issues, and translate requirements into POC and MVP features.

Research Assistant (ML Team Lead), SOLIDS Lab, UVIC (Victoria, BC) Sep 2021 - Aug 2024

- Developed **FUSTT**, a spatio-temporal transformer for multivariate long-term time-series forecasting, applying temporal feature engineering, sequence modeling, missing-data handling, model evaluation, & baseline comparison across real-world environmental datasets; sponsored by NRC, published in IEEE, & demonstrated lower forecasting error against Transformer, LSTM, & CNN-LSTM baselines.

- Applied neural networks & classical ML methods to environmental & sensor-based datasets, including forecasting, classification, EDA, error analysis, & model interpretation for research & decision-support use cases.
- Worked with domain experts & research collaborators to interpret model outputs, validate forecasting behavior, & connect predictive results to ecological monitoring & climate resilience use cases.
- Published peer-reviewed research in IEEE & Springer venues, filed/received 3 grants, & presented technical findings at CSCE 2022 & UVic/OCEAN CleanTech events.

SKILLS

Data Science & ML : Python, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, time-series forecasting, feature engineering, EDA, missing-data handling, model evaluation, regression, classification, statistical modeling, Random Forest, XGBoost, LSTM, Transformers, optimization fundamentals

Data Engineering & Analytics : T-SQL, SQLite3, PySpark, Microsoft Fabric, Azure Data Factory, Dataflow Gen2, ADLS Gen2, Power BI, Databricks, AWS S3, Cloudflare R2, Cloudflare D1, BigQuery, ETL/ELT pipelines, structured/semi-structured data processing

Production ML & System Design : FastAPI, REST APIs, Docker, Kubernetes, Git/GitHub Actions, release management, MLflow, modular ML workflows, CI/CD, Jenkins, Streamlit, model packaging, batch inference, API-based deployment, monitoring, tracing, production versioning, Bash/Shell

AI/LLM Systems & Decision Support : RAG, LangChain, LangSmith, Qdrant, ChromaDB, FAISS, OpenAI/, Ollama, GGUF, ONNX, Hugging Face, local/cloud LLM deployment, deterministic + LLM workflows, Edge AI deployment, business intelligence automation, Pydantic

Languages & Application Tools : Python, JavaScript, C/C++, SQL, KQL, OpenCV, Bash/Shell

EDUCATION

M.Eng. ECE – *University of Victoria* (2021–2024)

B.Eng. CSE – *Amity University* (2017–2021)

PUBLICATIONS

- **FUSTT**[†] – Spatio-temporal transformer for multivariate long-term time-series forecasting of marine stressors, sponsored by NRC & published through IEEE/NRC.
- **Concrete Characteristics Evaluation**[†] – Machine learning prediction of concrete properties for sustainability-focused material analysis & carbon-emission reduction.
- **Human Activity Recognition**[†] – Evaluated human activity detection using wearable sensor data & LSTM-based sequence modeling.

EXTERNAL PROJECTS

- **COMPIO**[†] – Business intelligence & financial profiling platform using Python, FastAPI, Yahoo Finance lookup, web/news scraping, RAG, S3 storage, & structured report generation for market intelligence workflows.
- **Fraud Detection ML**[†] – End-to-end classification pipeline using Logistic Regression & Random Forest, handling class imbalance & evaluating performance through confusion matrices, precision, recall, & F1-score.
- **Stock Forecasting**[†] – Time-series forecasting workflow comparing Linear Regression, LSTM, & XGBoost models with data preparation, feature engineering, prediction visualization, & app-based outputs.
- Additional projects across AI-assisted code workflow tools, local/cloud LLM applications, text-to-speech systems, computer vision, & production-oriented Python applications are available in my **portfolio & GitHub**.